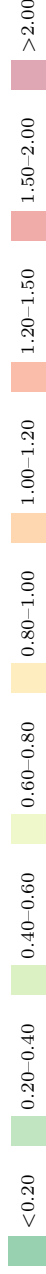


Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	Centaur-70B
Decision (2)	0.30	0.32	0.35	0.34	0.37	0.27	0.31	0.33	0.20	0.21	0.24	0.34	0.21	0.31	0.28	0.29	0.28	0.33
MDP (2)	0.56	0.56	0.58	<u>0.58</u>	<u>0.58</u>	0.48	0.53	0.57	0.39	0.46	0.52	0.57	0.42	0.57	0.51	0.53	0.53	0.50
Bandit (4)	0.32	0.35	0.37	0.36	<u>0.38</u>	0.33	0.33	0.36	0.24	0.23	0.32	0.34	0.28	0.34	0.22	0.25	0.25	0.39
Misc. (7)	0.28	0.31	0.45	0.46	<u>0.49</u>	0.14	0.30	0.40	0.03	0.11	0.18	0.36	0.09	0.35	0.37	0.41	0.29	0.52
Mean (15)	0.33	0.35	0.43	0.43	<u>0.45</u>	0.26	0.34	0.40	0.16	0.20	0.27	0.38	0.20	0.37	0.34	0.37	0.31	<u>0.46</u>
<0.05 0.05–0.12 0.12–0.18 0.18–0.25 0.25–0.32 0.32–0.40 0.40–0.48 0.48–0.55 ≥0.55																		

Table 1: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type on Psych-201 (out-of-distribution). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 15 experiments (of 18) with a well-defined discrete response space ($\ln k > 0$); 3 experiments with continuous or mixed responses are excluded. **Bold+underline** marks the best model, underline the second-best.

Experiment	Type	Queen3 family								Llama-3.1/3.2 family								[binz2025foundation]		Chance
		0.6B		1.7B		4B		8B		14B		1B		3B		8B				
		base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft			
Experience-description choice [anillo2024weird]	Decision	0.50	0.42	0.49	0.40	0.46	0.39	0.46	0.38	0.45	0.38	0.49	0.44	0.46	0.41	0.46	0.41	0.42	0.69	
Risky decision and happiness [rutledge2023happiness]	Decision	1.46	1.38	1.45	1.34	1.39	1.31	1.39	1.31	1.53	0.53	1.57	1.40	1.53	1.37	1.43	1.34	1.32	0.69	
Loss aversion task [spektor2024lossaversion]	Decision	0.57	0.54	0.56	0.54	0.54	0.52	0.53	0.54	0.53	0.50	0.66	0.57	0.61	0.55	0.54	0.52	0.51	0.69	
Two-step task [castrorodriguez2022twostep]	MDP	0.26	0.26	0.27	0.28	0.26	0.24	0.26	0.24	0.25	0.25	1.29	0.45	0.95	0.35	0.26	0.25	0.46	1.39	
Two-step task [shahar2019twosteptask]	MDP	0.55	0.48	0.53	0.47	0.54	0.47	0.55	0.47	0.53	0.46	0.54	0.49	0.53	0.47	0.53	0.47	0.46	0.69	
Range adaptation RL [bavard2021range]	Bandit	0.56	0.48	0.54	0.48	0.53	0.47	0.52	0.47	0.52	0.47	0.57	0.48	0.52	0.48	0.52	0.46	0.45	0.69	
Explore-exploit task [fan2022trait]	Bandit	0.42	0.28	0.40	0.27	0.40	0.26	0.39	0.27	0.40	0.26	0.42	0.28	0.43	0.27	0.39	0.26	0.26	0.69	
Counterfactual learning [palminteri2017confirmation]	Bandit	0.64	0.50	0.62	0.49	0.60	0.48	0.61	0.49	0.58	0.47	0.62	0.53	0.58	0.49	0.58	0.48	0.46	0.69	
Contextual reinforcement learning [vandendriessche2022depression]	Bandit	0.69	0.61	0.69	0.57	0.61	0.54	0.63	0.55	0.57	0.53	0.69	0.58	0.59	0.61	0.59	0.56	0.51	0.69	
Aversive reversal learning [zika2023traitanxiety]	Bandit	3.66	3.36	3.70	3.36	3.57	3.29	3.51	3.27	3.47	3.27	4.00	3.99	3.90	3.85	3.87	3.82	3.80	—	
Navon task [busch2024navon]	Misc.	0.43	0.40	0.40	0.35	0.43	0.32	0.41	0.32	0.37	0.32	0.47	0.59	0.44	0.45	0.39	0.34	0.31	0.69	
Stroop task [busch2024stroop]	Misc.	0.53	0.38	0.28	0.63	0.21	0.18	0.20	0.17	0.21	0.15	0.97	0.85	0.29	0.45	0.20	0.18	0.15	1.10	
Pragmatic language task [franke2024bayesian]	Misc.	1.55	1.25	1.47	1.03	1.17	0.81	1.08	0.76	1.07	0.77	1.64	1.49	1.75	1.25	1.85	1.07	0.75	1.39	
Reference game [frankedegen2016reasoning]	Misc.	1.50	1.21	1.33	1.12	1.04	0.83	1.14	0.85	0.97	0.80	1.47	1.34	1.41	1.12	1.19	1.04	0.80	1.39	
Compound word processing [guenther2020ts]	Misc.	0.62	0.49	0.52	0.41	0.37	0.37	0.36	0.37	0.34	0.33	0.49	0.47	0.42	0.38	0.37	0.34	0.32	0.69	
Grammaticality judgement [guenther2023grammaticality]	Misc.	0.47	0.43	0.37	0.41	0.35	0.36	0.35	0.31	0.32	0.30	0.50	0.42	0.40	0.39	0.34	0.33	0.29	0.69	
Scalar inference [tsvilodub2023xorsome]	Misc.	3.37	3.33	3.29	3.19	3.13	3.04	3.06	3.00	2.97	2.95	3.62	3.50	3.48	3.38	3.23	3.20	2.91	—	
Math arithmetic task [xu2023augmenting]	Misc.	0.72	0.72	0.71	0.71	0.71	0.70	0.69	0.70	0.68	0.66	0.73	0.72	0.73	0.72	0.71	0.72	0.55	0.69	
Mean (18)		1.03	0.92	0.98	0.89	0.91	0.81	0.90	0.80	0.87	0.79	1.15	1.03	1.06	0.94	0.97	0.88	0.82	0.86	



Bold+underline = best; underline = second-best. Lower is better. “—” = continuous or mixed response space. Size labels in billions of parameters.

Table 2: Per-experiment NLL on Psych-201 (out-of-distribution). Psych-201 was not used during supervised fine-tuning; lower NLL indicates better generalisation. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. The ln(*k*) column shows the random-guessing baseline where *k* is the number of per-trial response options.

Experiment	Type	Qwen2.5				Llama-3.3				SmoLLM				Ollama				Chance
		0.5B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	70B	ln(k)	
Experience-description choice [anillo2024weird]	Decision	0.42	0.40	0.39	<u>0.38</u>	0.38	0.44	0.41	0.41	0.53	0.52	0.47	0.40	0.49	0.41	0.42	0.69	
Risky decision and happiness [rutledge2023happiness]	Decision	1.38	1.40	1.34	<u>1.31</u>	1.33	1.40	1.37	1.34	1.50	1.44	1.46	1.36	1.48	1.38	1.32	—	
Loss aversion task [spektor2024lossaversion]	Decision	0.54	0.54	0.52	0.54	0.50	0.57	0.55	0.52	0.58	0.58	0.58	0.52	0.61	0.54	0.51	0.69	
Two-step task [castro-rodriguez2022twostep]	MDP	0.26	0.28	<u>0.24</u>	0.24	0.25	0.45	0.35	0.25	0.48	0.43	0.34	0.25	0.47	0.26	0.46	1.39	
Two-step task [shahar2019twostep]	MDP	0.48	0.47	0.47	0.47	0.46	0.49	0.47	0.47	0.61	0.54	0.50	0.47	0.58	0.47	0.46	0.69	
Range adaptation RL [bavard2021range]	Bandit	0.48	0.48	0.47	0.47	0.47	0.48	0.48	0.46	0.54	0.55	0.51	0.48	0.55	0.48	0.45	0.69	
Explore-exploit task [fan2022trait]	Bandit	0.28	0.27	0.26	0.27	0.26	0.28	0.27	0.26	0.35	0.32	0.29	0.27	0.29	0.27	0.26	0.69	
Counterfactual learning [palminteri2017confirmation]	Bandit	0.50	0.49	0.48	0.49	0.47	0.53	0.49	0.48	0.59	0.60	0.52	0.52	0.59	0.51	0.46	0.69	
Contextual reinforcement learning [vandendriesche2022depression]	Bandit	0.61	0.57	0.54	0.55	<u>0.53</u>	0.58	0.61	0.56	0.64	0.68	0.58	0.57	0.57	0.58	0.51	0.69	
Aversive reversal learning [zika2023traitanxiety]	Bandit	3.36	3.36	3.29	3.27	3.27	3.99	3.85	3.82	3.54	3.44	3.33	3.89	4.35	4.00	3.80	—	
Navon task [busch2024navon]	Misc.	0.40	0.35	<u>0.32</u>	0.32	0.32	0.59	0.45	0.34	0.65	0.61	0.44	0.34	0.72	0.35	0.31	0.69	
Stroop task [busch2024stroop]	Misc.	0.38	0.63	0.18	0.17	<u>0.15</u>	0.85	0.45	0.18	0.67	0.76	0.68	0.29	0.88	0.39	0.15	1.10	
Pragmatic language task [franke2024bayesian]	Misc.	1.25	1.03	0.81	0.76	0.77	1.49	1.25	1.07	1.61	1.64	1.54	1.06	1.39	1.08	0.75	1.39	
Reference game [frankel2016reasoning]	Misc.	1.21	1.12	0.83	0.85	0.80	1.34	1.12	1.04	1.92	1.35	1.45	1.14	1.38	1.04	0.80	1.39	
Compound word processing [gunther2020ts]	Misc.	0.49	0.41	0.37	0.37	0.33	0.47	0.38	0.34	0.61	0.52	0.44	0.35	0.56	0.42	0.32	0.69	
Grammaticality judgement [gunther2023grammaticality]	Misc.	0.43	0.41	0.36	0.31	<u>0.30</u>	0.42	0.39	0.33	0.49	0.45	0.42	0.40	0.45	0.35	0.29	0.69	
Scalar inference [tsvilodub2023xorsome]	Misc.	3.33	3.19	3.04	3.00	<u>2.95</u>	3.50	3.38	3.20	3.78	3.60	3.45	3.28	3.96	3.30	2.91	—	
Math arithmetic task [xu2023augmenting]	Misc.	0.72	0.71	0.70	0.70	<u>0.66</u>	0.72	0.72	0.72	0.78	0.75	0.74	0.72	0.75	0.72	0.55	0.69	
Mean (18)		0.92	0.89	0.81	0.80	0.79	1.03	0.94	0.88	1.10	1.04	0.99	0.91	1.11	0.92	0.82	0.86	

Bold+underline = best; underline = second-best (across the four families and Centaur-70B). Lower is better. "—" = continuous or mixed response space. Size labels in billions of parameters. 70B is the reproduced Centaur-70B.

Table 3: Per-experiment NLL on Psych-201 (out-of-distribution) for the four finetuned families. Psych-201 was not used during fine-tuning; lower NLL indicates better generalisation. The ln(k) column shows the random-guessing baseline where k is the number of per-trial response options.